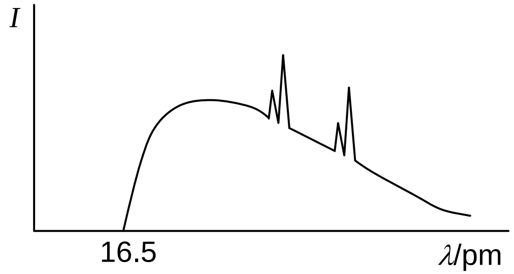
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)	Background spectra as skewed normal curve and line spectra shown (1) Minimum wavelength labelled /shown as 1.65×10^{-11} m (1) e.g. 	1	1		2	1	
		(ii)	Power = 9 000 W (1) Rate of heat = 8 937 W (1)		2		2	2	
		(iii)	Rearrangement $\mu = \dfrac{\ln\left(\dfrac{100}{60}\right)}{4.1\times10^{-3}}$ **OR** 0.6 × 0.6 (1) μ = 365 m^{-1} answer with unit **OR** 0.36 (1) Substitution into $I = I_0\,e^{-\mu x}$ where $x = 2.8 \times 10^{-3}$ so I = 36% (1)		3		3	3	
	(b)	(i)	Ultrasound hits blood cells (1) Reflected pulse different frequency/wavelength (1)	2			2		
		(ii)	Rearrangement $v = \dfrac{\Delta f c}{2 f_0 \cos\theta}$ (1) v = 0.098 [m/s] (1) Answer of 0.18 [m/s] award 1 mark only for the question part		2		2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	**Any 3 of 4** Comment regarding half-life e.g. long enough to enable measurement / short so effect soon disappears (1) Gamma emitter so can be detected out of the body (1) Daughter product stable (1) Low ionisation (1)			3	3		
		(ii)	Gamma rays pass through a <u>collimator</u> (1) Cause scintillations / light flashes on a crystal (1) Collected by a series of photomultiplier tubes OR CCD (1)	3			3		
		(iii)	Mass-energy of electron [or positron] should be equal to the photon energy [because $2e \rightarrow 2\gamma$] (1) $E_{electron} = mc^2 = 8.20 \times 10^{-14}$ J (1) $\therefore E_{ph} = \frac{8.20\times10^{-14}\,[\text{J}]}{1.6\times10^{-13}\,[\text{J MeV}^{-1}]}$ = 0.512 MeV		1	1 1	3	2	
			Question 7 total	**6**	**9**	**5**	**20**	**10**	**0**